CSC4197

RESEARCH METHODOLOGY

LITERATURE REVIEW & RESEARCH GAP IDENTIFICATION

2 WHY REVIEW LITERATURE AT ALL?

- Verify **novelty** — has your problem been solved?
- Learn the **state of the art** (methods, datasets, baselines, metrics)
- Find the **gap** your work will fill
- Borrow **methodology** — stand on the shoulders of giants
- Avoid **reinventing** (or worse, unknowingly plagiarizing) prior work
- A paper without a related-work grounding is the #1 desk-reject reason at good venues.

3 TYPES OF LITERATURE REVIEW

Type	Purpose	Rigor
Narrative	Background overview for a paper/thesis intro	Informal
Scoping	Map the breadth of a new area	Semi-formal
Systematic (SLR)	Answer a defined question via exhaustive protocol	Formal (PRISMA)
Meta-analysis	Statistically combine results of many studies	Most formal

For your mini-paper: **narrative** review of 8–10 strong papers. Thesis students: consider an SLR chapter.

4 PRISMA IN 60 SECONDS (SYSTEMATIC REVIEWS)

- The PRISMA flow for an SLR:
  - **Identification** — search N databases with a documented query → X records
  - **Screening** — remove duplicates; screen titles/abstracts against inclusion/exclusion criteria
  - **Eligibility** — full-text review of survivors
  - **Included** — final set analyzed
- Everything is **logged and reproducible** — another team running your protocol should get the same paper set.

5 SEARCH STRATEGY: DATABASES & OPERATORS

- **Where to search (CS):** IEEE Xplore, ACM Digital Library, Scopus, Web of Science, SpringerLink, ScienceDirect, arXiv (preprints), Google Scholar (broad net)
- **Query craft:**
- Boolean: `("intrusion detection" AND "deep learning") NOT survey`
- Wildcards: `optimi*` → optimize/optimise/optimization
- Field limits: title/abstract only; year ≥ 2020 for fast-moving areas
- **Snowballing:** backward (their references) + forward ("cited by")
- **Demo (live):** building a query in IEEE Xplore via AIUB library access

6 EVALUATING SOURCE QUALITY

- **Peer-reviewed?** Journals & ranked conferences > random PDFs
- **Venue rank:** CORE (A*/A/B/C) for CS conferences; Scimago Q1–Q4 / JCR impact factor for journals
- **Citations** (age-adjusted) — a 2025 paper with 30 cites ≫ a 2010 paper with 30
- ⚠️ **Predatory journals:** pay-to-publish, fake review, weeks-long "peer review," flattering spam emails
  - Check: Think-Check-Submit, journal's indexing claims, editorial board reality
- arXiv preprints: usable in fast fields (ML), but verify if later peer-reviewed

7 THEORY: TAXONOMY OF RESEARCH GAPS

- **Evidence gap** — contradictory results across studies
- **Knowledge gap** — phenomenon simply unstudied
- **Methodological gap** — studied, but with limited methods (e.g., only simulations, never real users)
- **Population/context gap** — studied elsewhere, not for your population (e.g., low-resource languages, Bangladeshi datasets)
- **Theoretical gap** — results exist but lack an explaining framework
- Your related-work section must **name** the gap explicitly: "*However, no prior work has...*"

8 THE LITERATURE SYNTHESIS MATRIX

Build a table — one row per paper:

Paper (Year)	Problem	Method	Dataset	Results	Limitation (→ gap!)
Rahman et al. 2024	Bangla sentiment	BiLSTM	10k tweets	F1 0.81	No code-mixed text
Das & Khan 2025	Code-mixed sentiment	mBERT	5k posts	F1 0.77	Small data; no LLM baseline

The **Limitation column read vertically = your gap**

9 READING WORKSHOP: THE THREE-PASS METHOD (KESHAV)

- **Pass 1 (5–10 min):** title, abstract, intro, headings, conclusion → *category, context, correctness, contributions, clarity*
- **Pass 2 (~1 hr):** read fully but skip proofs; study figures/tables; note unread references
- **Pass 3 (1–4 hrs):** virtually re-implement — challenge every assumption
- **In class now (25 min):** Pass 1 + partial Pass 2 on the provided paper → fill one synthesis-matrix row individually, then compare with your neighbor.

10 CASE STUDY: FINDING A GAP IN 8 PAPERS

- **Mini-case:** A group reviewing "ML for dengue outbreak prediction in South Asia" finds:
- 5 papers use Indian climate data; 2 use Sri Lankan; 1 uses Dhaka data **from 2015**
- All use classical ML (RF/SVM); none compare against modern temporal models
- None release code
- **Gaps identified:** population gap (recent Bangladeshi data), methodological gap (no temporal deep learning), reproducibility gap.
- **Resulting RQ (preview of Week 4):** "How accurately can temporal deep models predict dengue cases in Dhaka using 2020–2025 climate + case data, compared to classical baselines?"

11 TOOLS FOR LITERATURE DISCOVERY

- **Connected Papers** — visual graph of related work from one seed paper
- **Litmaps / Research Rabbit** — citation-network exploration & alerts
- **Semantic Scholar** — TLDRs, influential-citation counts
- **Google Scholar alerts** — auto-email new papers matching your query
- (Week 9 covers *managing* what you find — Zotero; Week 10 covers AI tools like Elicit responsibly)
- **Hands-on (10 min):** seed Connected Papers with one paper from your homework; screenshot the graph; identify 2 new must-read papers.

12 WRAP-UP, ASSIGNMENT 1 & NEXT WEEK

- **Key takeaways**
- A literature review is an argument, not a list of summaries
- Quality-filter sources; log your search; synthesize in a matrix
- The gap must be named and typed (evidence/knowledge/method/population/theory)
- **Homework:** 8–10 paper synthesis matrix + 1-page narrative review + explicit gap statement for your group's problem (from Week 2)
- **Next week:** Turning gaps into research questions & hypotheses